BEYOND HEMISPHERIC LATERALITY: RIEMANNIAN METRICS AS MARKERS OF MOTOR IMAGERY PERFORMANCE IN BCI TRAINING

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1 Introduction

Background & The BCI Illiteracy Problem

- Motor imagery (MI)-based BCIs enable control through imagined movements.
- BCI illiteracy: MI-BCI performance varies substantially across users, with ~10–30% failing to achieve reliable control.
- Successful MI-BCI control requires neural patterns that are stable and discriminable.
- Traditional evaluation focuses on classifier accuracy which is explained mainly by hemispheric lateralization (left–right class separability)
- Riemannian geometry-based metrics provide a classifier-independent approach to quantify three dimensions of MI-BCI performance (Lotte and Jeunet 2018):
 - Class distinctiveness → separation between imagery classes
 - Class stability → consistency within a class
 - Rest distinctiveness → separation from baseline activity

3 Aim

To investigate how Riemannian neural metrics evolve during MI-BCI training and how their changes relate to users' learning trajectories.

2 Methodology



Dataset:

Source: Cho et al. (2017), 64-channel Biosemi ActiveTwo EEG (512 Hz),
Paradigm: Left vs. Right Hand Motor Imagery (MI),
Volume: 5-6 Runs per subject, 100-120 trials per class.
Subjects: 52 (Reduced to N=44 after QC).
Reference: Common Average Reference (CAR).

→ **Cohort Selection:** N = 44 (Strict exclusion of 8 subjects for artifacts/missing data).

→ **Preprocessing Pipeline:**
Filtering: 0.1–50 Hz Bandpass + 60 Hz Notch.
Artifact Removal: ASR (Transient noise) + ICA (Infomax) for ocular/myogenic artifacts.
Epoching: Cropped to 0s–3s post-stimulus (active imagery).

1. Riemannian Neural Metrics
- Class Distinctiveness: Left ↔ Right MI
- Rest Distinctiveness: MI ↔ Baseline
- Class Stability: Within-class consistency

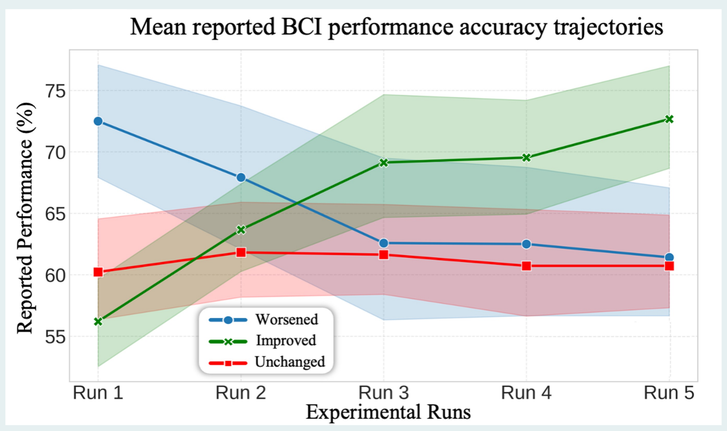
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2. Behavioral Learning Trajectories
Expected BCI performance across 5 runs
- Improved ↗
- Unchanged →
- Worsened ↘

3. Run-wise Neural Changes
OLS slope across 5 runs

CORRELATION ANALYSIS	GROUP DIFFERENCES	OVR SIGNATURES
Neural metrics ↔ Expected BCI performance Spearman rank correlation Across frequency bands & groups	Neural metric slopes across groups Kruskal–Wallis + 5,000-permutation + post-hoc Mann–Whitney U + Holm correction	Target group vs remaining participants Cohen's d on neural metric slopes Group-specific neural trends

4 Results



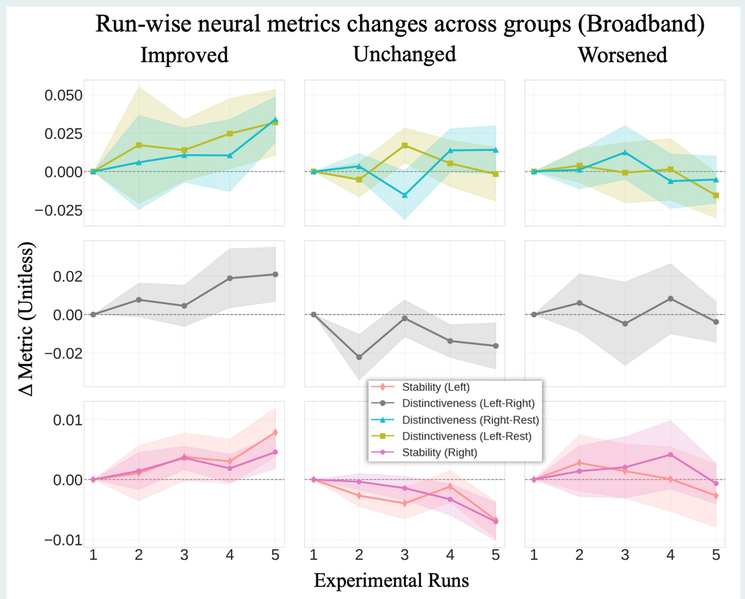
1. Behavioral Stratification by Expected Slope

• Improved (n = 15): Learners who steadily improved from ~56% to >72% accuracy.

• Unchanged (n = 16): Stable plateau with stagnant performance at ~60–62%.

• Worsened (n = 13): Participants whose performance declined from ~72% down to ~61%

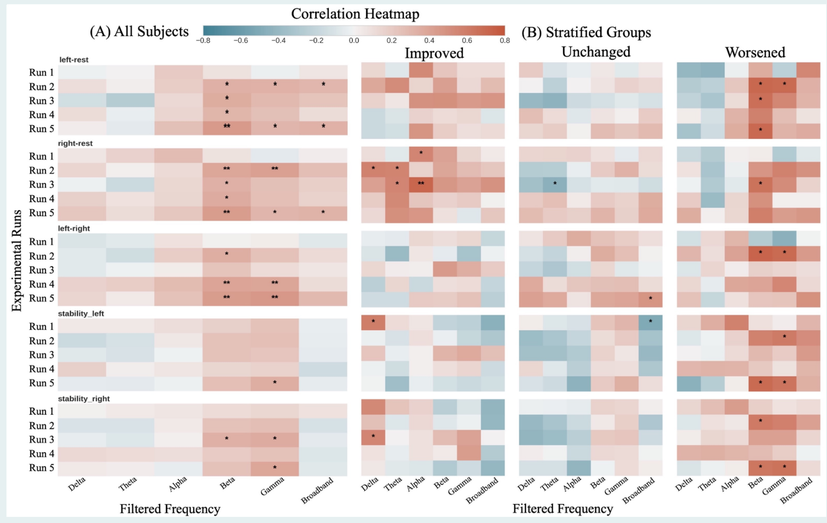
Inference: MI-BCI performance showed heterogeneous learning trajectories across participants.



3. Run-wise Neural Metric Changes Across Learning Groups

- Improved: Increasing rest distinctiveness, class stability, and left–right distinctiveness across runs.
- Unchanged: Decrease in class stability and left–right distinctiveness.
- Worsened: No change in left–right distinctiveness and stability, but decreasing rest distinctiveness.

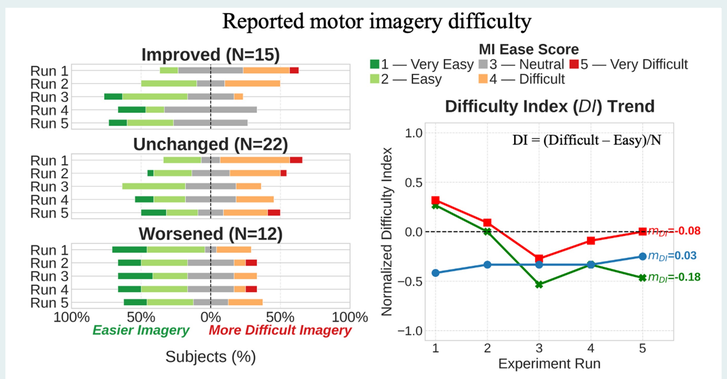
Inference: Neural representations evolve differently across learning trajectories, with no single metric consistently explaining performance changes.



4. Frequency-Specific Neural Correlates of MI-BCI Performance

- All subjects: Stronger correlations were observed particularly in the beta and gamma bands, especially for rest distinctiveness and left–right separability.
- Improved: Showed stronger and more consistent correlations in lower frequency bands.
- Unchanged: Showed weaker and more heterogeneous relationships.
- Worsened: Correlations were primarily observed in higher frequency bands.

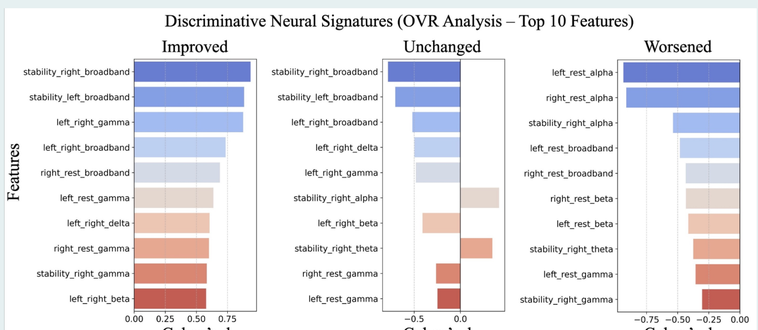
Inference: Behavioral stratification reveals distinct frequency-specific relationships between neural metrics and expected MI-BCI performance.



2. Motor Imagery Difficulty Across Learning Trajectories

- Improved: Moderate imagery difficulty, with a temporary reduction around Run 3.
- Unchanged: Consistently higher perceived imagery difficulty across runs.
- Worsened: Relatively easy motor imagery throughout training.

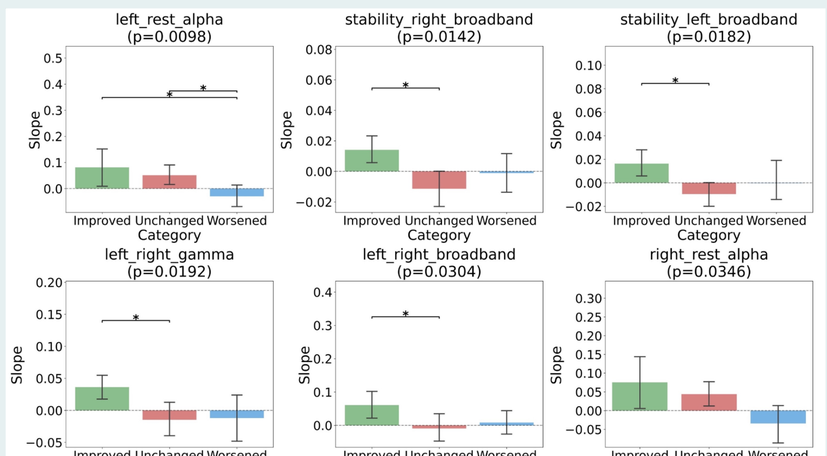
Inference: Perceived imagery difficulty did not directly correspond to expected BCI performance.



5. One-vs-Rest (OVR) Neural Signatures

- Improved: Strong positive effect sizes for broadband stability and left–right separability.
- Unchanged: Negative effect sizes for stability and class-separability metrics, despite increased task–rest divergence.
- Worsened: Large negative effect sizes for rest-related alpha and broadband metrics.

Inference: Each learning trajectory exhibited a distinct neural signature across multiple representational dimensions.



6. Divergent Neural Adaptation Trends Across Behavioral Groups

- Improved: Positive slopes in alpha-band rest distinctiveness and increasing broadband stability and left–right distinctiveness.
- Unchanged: Broadband stability decreased, while left–right distinctiveness showed limited change
- Worsened: Alpha-band rest distinctiveness decreased, while other separability measures remained stable or decreased.

Inference: Improved participants showed positive neural trends across multiple representational dimensions.

5 Conclusion

MI-BCI learning cannot be explained by hemispheric lateralization alone.

- Successful learning is associated with the coordinated evolution of:
 - Class stability
 - Class distinctiveness
 - Rest distinctiveness
- Riemannian neural metrics provide a multidimensional characterization of MI-BCI learning trajectories.

6 References

- Lotte, F., & Jeunet, C. (2018). Defining and quantifying users' mental imagery-based BCI skills: A first step. Journal of Neural Engineering, 15(4), 046030.
- Cho, H., Ahn, M., Ahn, S., Kwon, M., & Jun, S. C. (2017). EEG datasets for motor imagery brain-computer interface. GigaScience, 6(7), gix034.

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